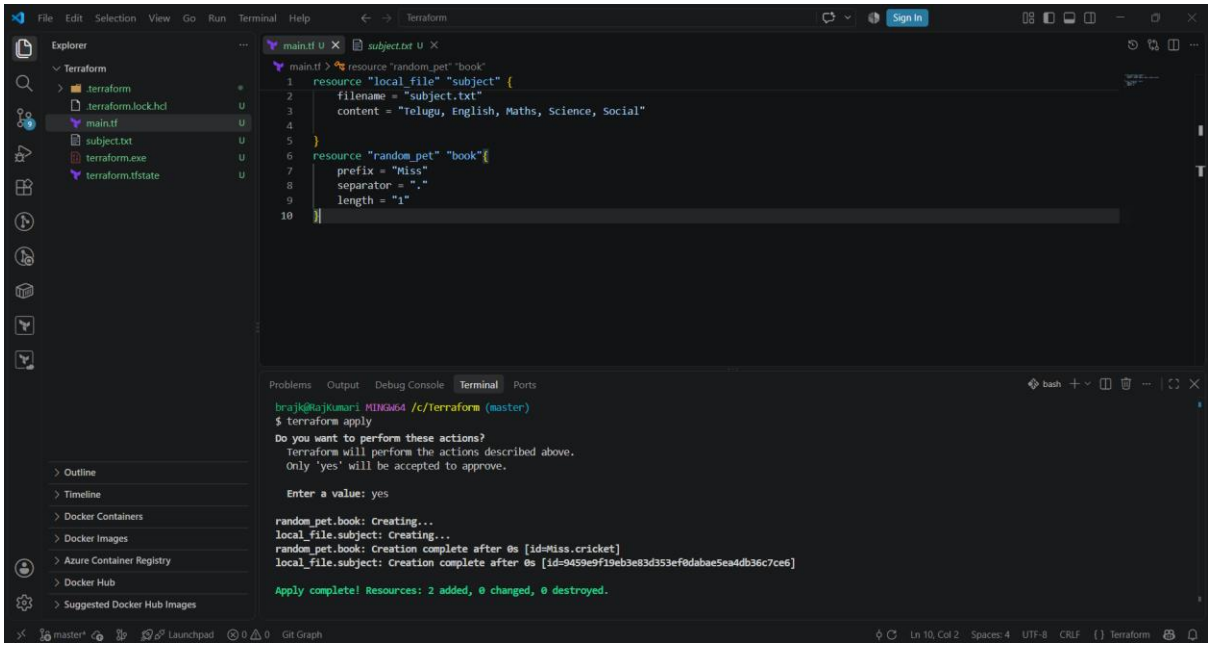
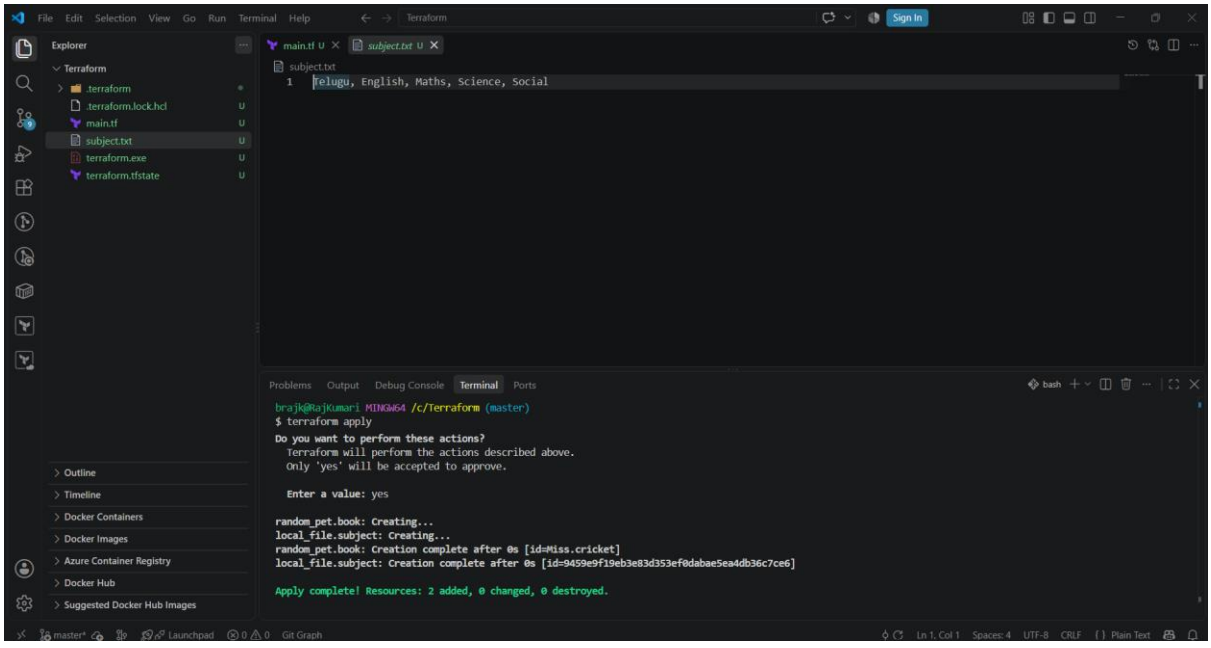
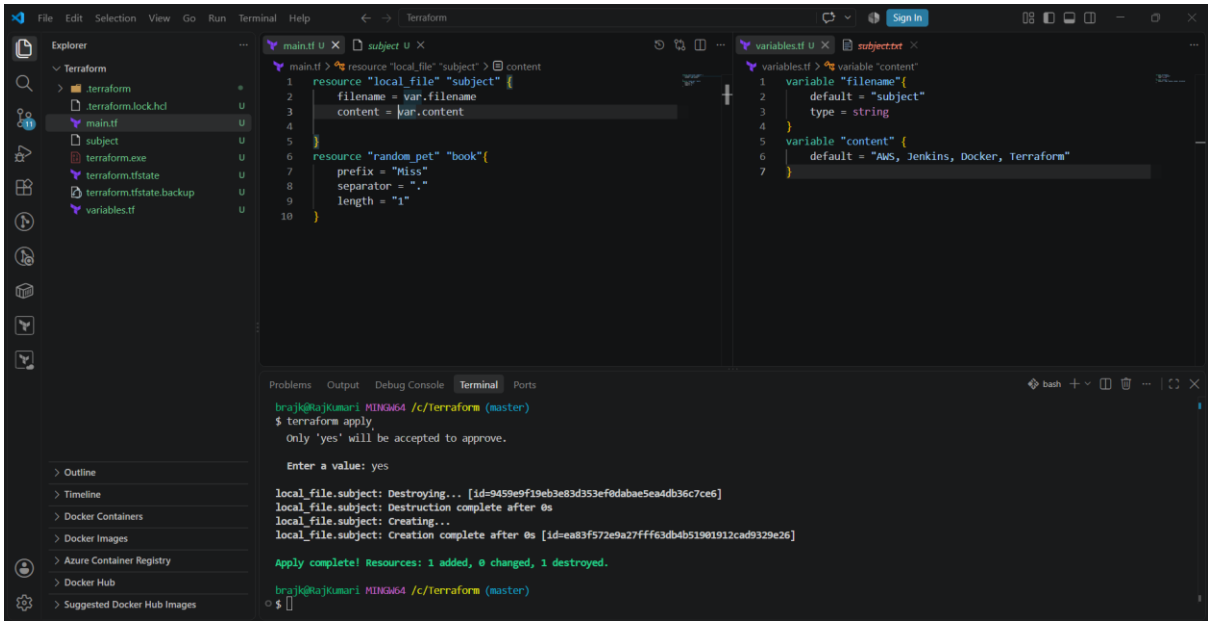
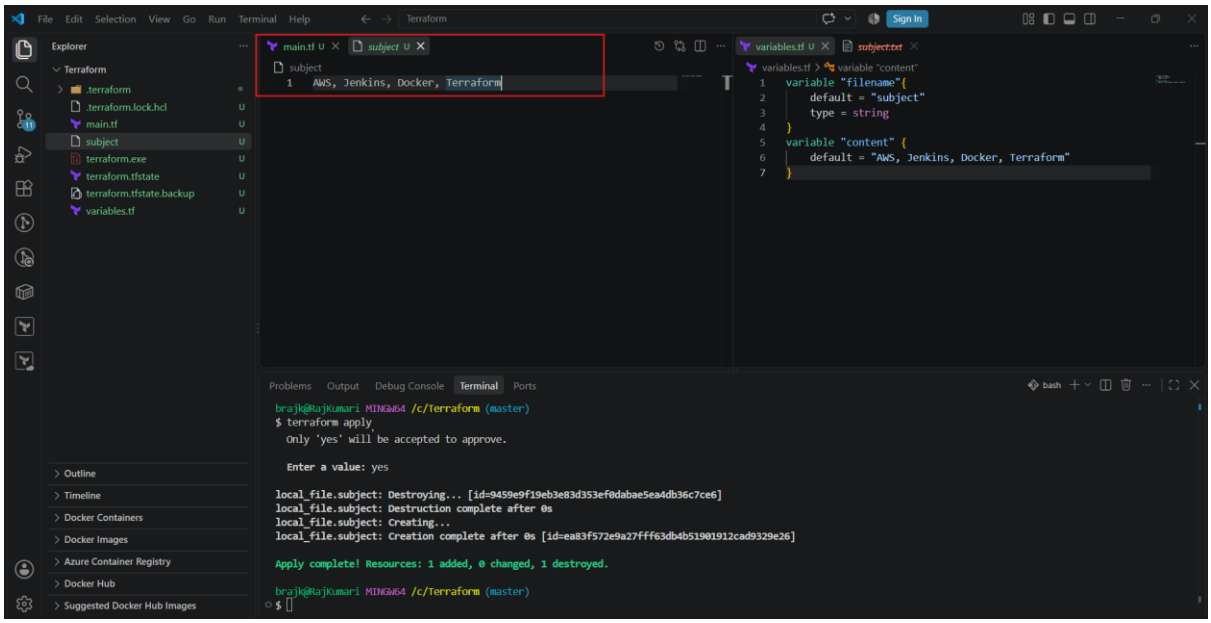
TERRAFORM TASK - 03

1.Execute the script shown in the video.






File Edit Selection View Go Run Terminal Help
Terraform
Sign In
Explorer
Terraform
.terraform
.terraform.lock.hcl
main.tf
subject
terraform.exe
terraform.tfstate
terraform.tfstate.backup
variables.tf
main.tf U
subject U
main.tf > resource "local_file" "subject" > content
resource "local_file" "subject" {
filename = var.filename
content = var.content
}
resource "random_pet" "book"{
prefix = "Miss"
separator = "."
length = "1"
}
variables.tf U
subject.txt
variables.tf > variable "content"
variable "filename"{
default = "subject"
type = string
}
variable "content" {
default = "AWS, Jenkins, Docker, Terraform"
}
Problems Output Debug Console Terminal Ports
bash
brajk@RajKumari MINGW64 /c/Terraform (master)
$ terraform apply
Only 'yes' will be accepted to approve.
Enter a value: yes
local_file.subject: Destroying... [id=9459e9f19eb3e83d353ef0dabae5ea4db36c7ce6]
local_file.subject: Destruction complete after 0s
local_file.subject: Creating...
local_file.subject: Creation complete after 0s [id=ea83f572e9a27fff63db4b51901912cad9329e26]
Apply complete! Resources: 1 added, 0 changed, 1 destroyed.
brajk@RajKumari MINGW64 /c/Terraform (master)
$
Outline
Timeline
Docker Containers
Docker Images
Azure Container Registry
Docker Hub
Suggested Docker Hub Images


File Edit Selection View Go Run Terminal Help
Terraform
Sign In
Explorer
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.terraform
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main.tf
subject
terraform.exe
terraform.tfstate
terraform.tfstate.backup
variables.tf
main.tf U
subject U
subject
1 AWS, Jenkins, Docker, Terraform
variables.tf U
subject.txt
variables.tf > variable "content"
variable "filename"{
default = "subject"
type = string
}
variable "content" {
default = "AWS, Jenkins, Docker, Terraform"
}
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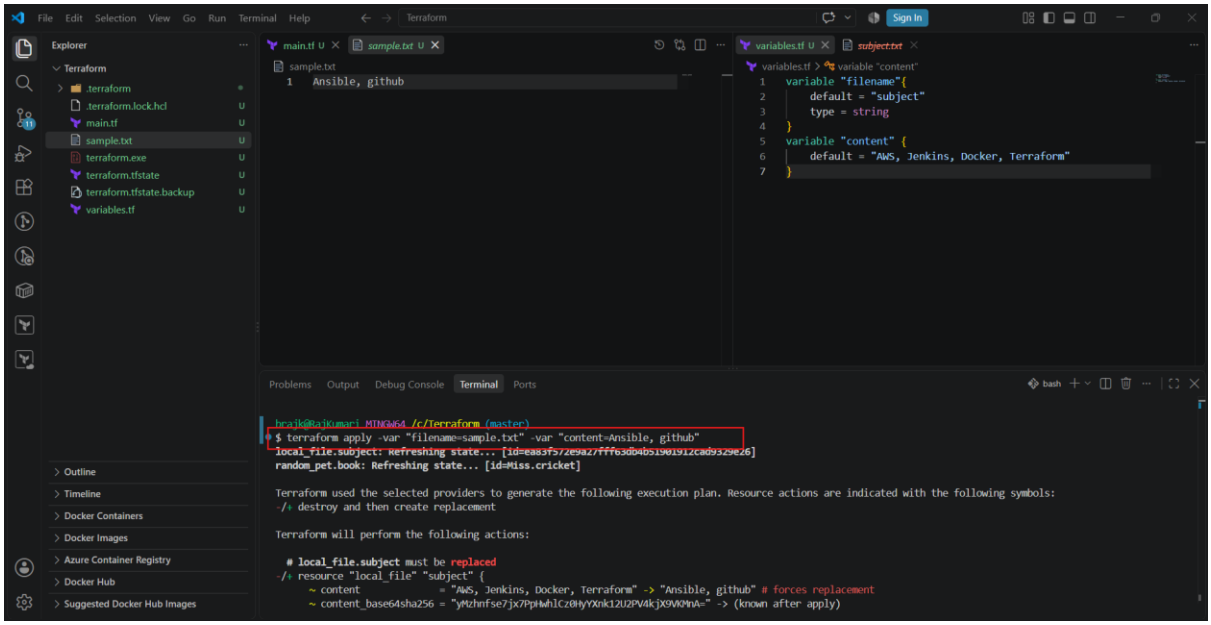
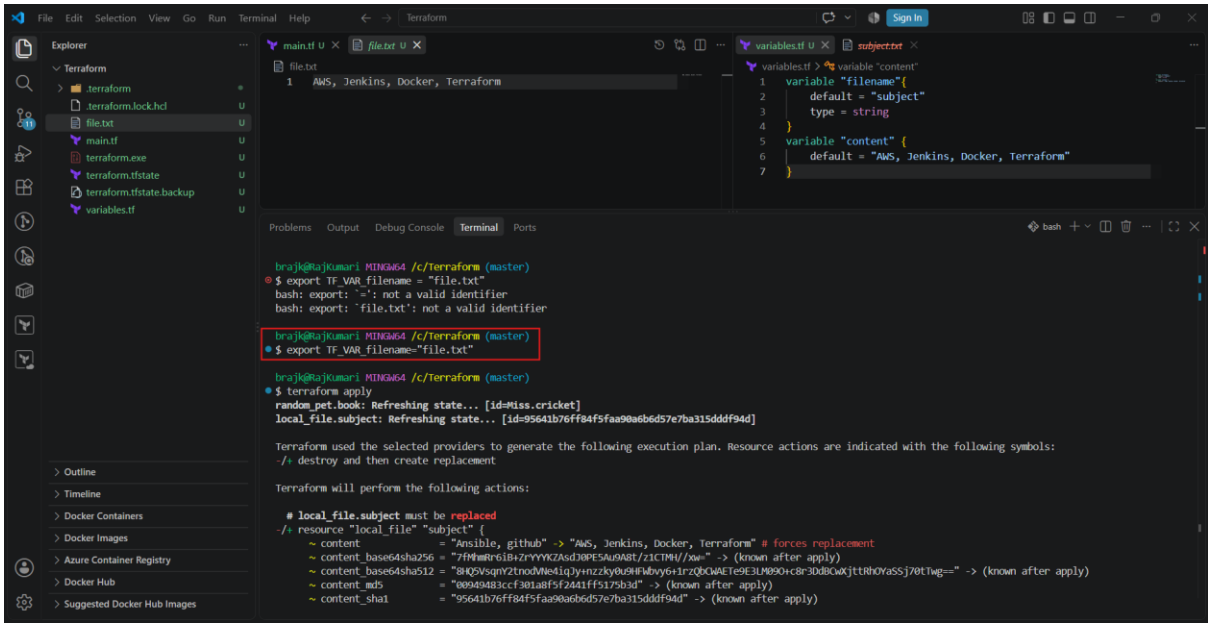


Environmental Variables in Terraform: Terraform environment variables are variables defined at the operating-system level and automatically consumed by Terraform using **the TF_VAR_ prefix to provide values** for Terraform input variables without hard-coding them in **.tf files.**

Why use them?

- Avoid hard-coding configuration values
- Different values can be used for different environments
- Useful in CI/CD pipelines such as Jenkins
- Can help keep sensitive values out of .tf files (though secrets should still be handled carefully)

Example: **export TF_VAR_filename="file.txt"**



Terraform Output Variable: A **Terraform output variable** is used to **display or expose information about resources created by Terraform** after terraform apply.

In simple words: **Output variables allow Terraform to show important values from your infrastructure, such as an EC2 instance IP address, DNS name, VPC ID, or S3 bucket name.**

Example

Suppose Terraform creates an EC2 instance:

```
resource "aws_instance" "server" {
  ami           = "ami-xxxxxxxx"
  instance_type = "t2.micro"
}
```

You can create an output variable:

```
output "instance_public_ip" {
  description = "Public IP address of the EC2 instance"
  value       = aws_instance.server.public_ip
}
```

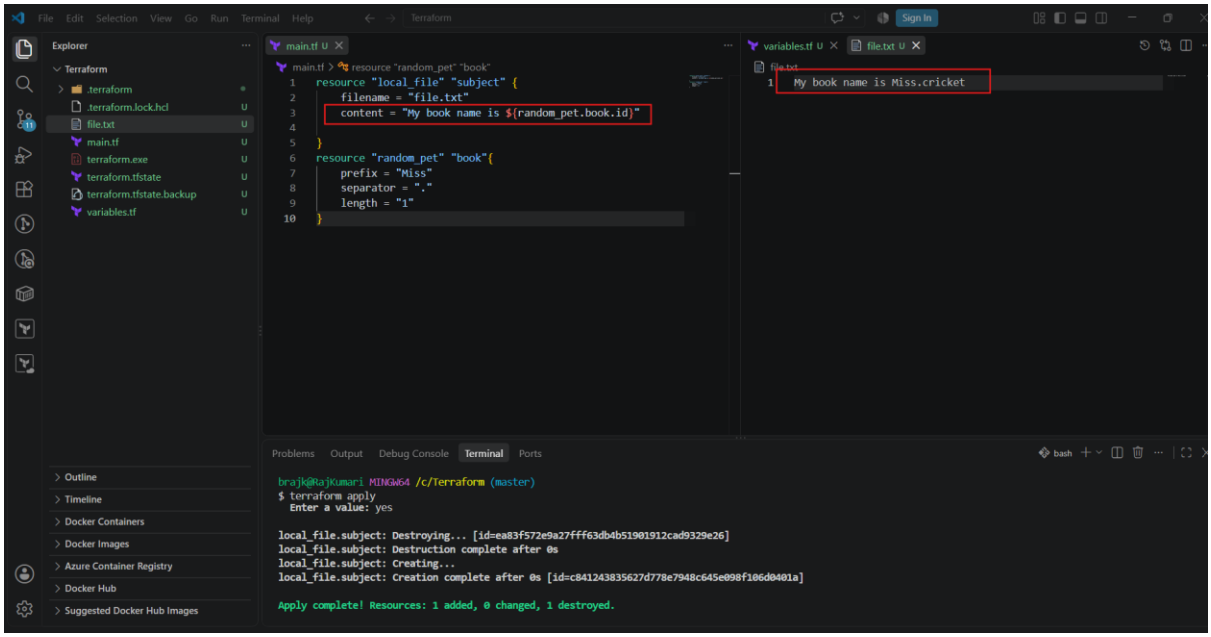
After running:

terraform apply

Terraform displays something like:

Apply complete! Resources: 1 added, 0 changed, 0 destroyed.

Outputs: instance_public_ip = "54.123.45.67"



2.Integrate Terraform in Jenkins using the Terraform plugin.

GitHub → Jenkins → Terraform Plugin → Terraform → AWS

One important point first: the Jenkins **Terraform plugin** is an older plugin (current listed version 1.0.10), and its documented configuration is primarily a **build wrapper**. For a modern Jenkins pipeline, you can still configure Terraform through **Manage Jenkins → Tools** and invoke Terraform from a Jenkinsfile. This is generally easier to maintain.

1. Overall architecture

Developer

|

| git push

v

GitHub Repository

|

v

Jenkins

|

| Terraform Plugin / Terraform installation

v

Terraform

|

| AWS credentials

v

AWS

|

+---- EC2

+---- VPC

+---- Security Group

+---- S3

+---- etc.

The important distinction is:

- **Jenkins Terraform plugin** → helps Jenkins integrate with Terraform.
- **Terraform AWS provider** → allows Terraform to create/manage AWS resources.

Terraform itself uses provider plugins to communicate with services such as AWS.

Step 2: Install Terraform on the Jenkins server

SSH into your Jenkins EC2 instance.

Check whether Terraform is already installed:

terraform version

If you get something like:

Terraform v1.x.x

you can proceed.

If not, install Terraform.

For example, on Amazon Linux:

sudo dnf update -y

Then install Terraform using HashiCorp's repository/package instructions appropriate for your Amazon Linux version.

After installation:

terraform version

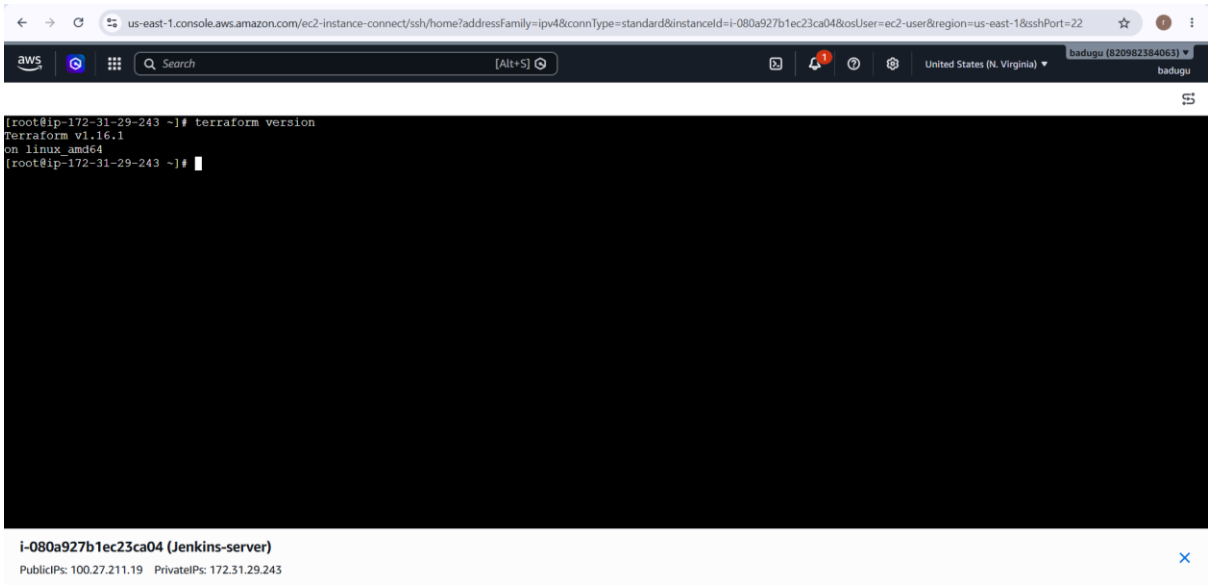
You should see:

Terraform v1.x.x

on linux_amd64

Why this is important

Jenkins ultimately needs access to the terraform executable on the machine/agent where the build runs.



Step 3: Install the Jenkins Terraform plugin

Go to:

Jenkins Dashboard

↓

Manage Jenkins

↓

Plugins

↓

Available plugins

Search:

Terraform

Install:

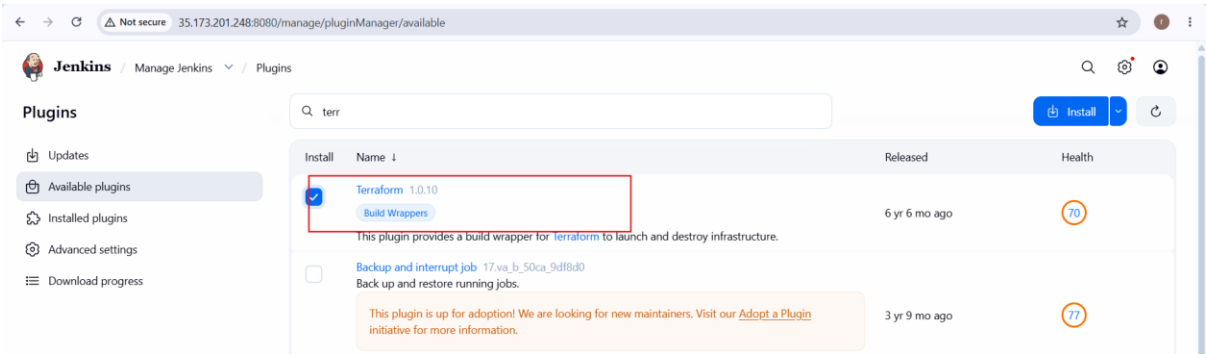
Terraform

The official Jenkins plugin page identifies the plugin ID as:

terraform

and documents installation through the Jenkins plugin manager/CLI.

After installation, restart Jenkins if Jenkins asks you to.



Step 4: Configure Terraform in Jenkins

Go to:

Jenkins Dashboard

↓

Manage Jenkins

↓

Tools

Look for:

Terraform installations

Click:

Add Terraform

Give it a name, for example:

terraform

or:

Terraform-1.13

You can configure the Terraform executable/installation here.

For example:

Name:

terraform

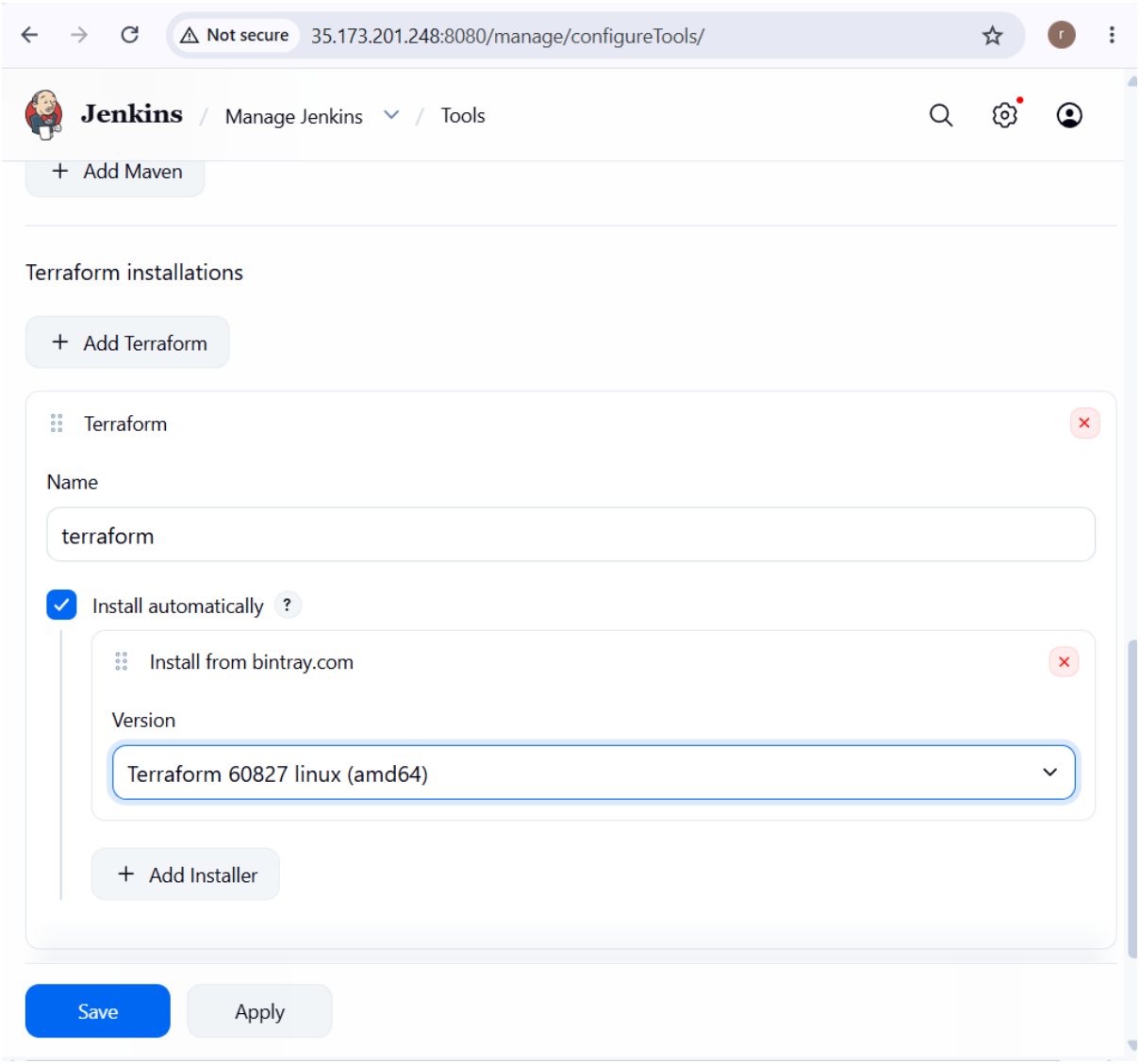
Install automatically:

✓

Or, if Terraform is already installed on your Jenkins server, configure Jenkins to use the existing Terraform installation.

Then click:

Save



3.Create one Jenkins job using Maven Project for the code below with two stages:

- **Stage 1: Git clone**

- **Stage 2: Maven Compilation Code: https://github.com/betawins/java-Working-app.git**

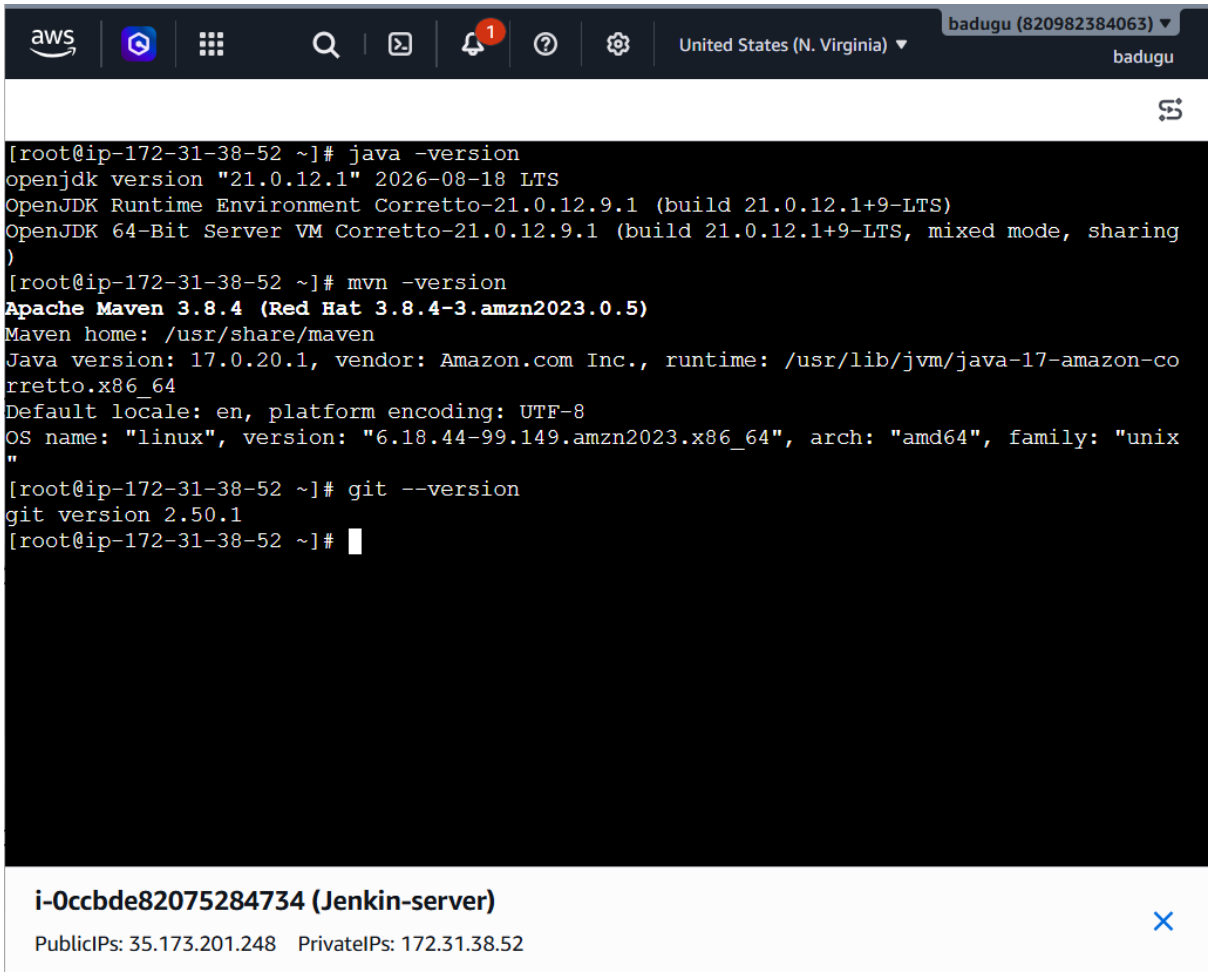
1. First verify Maven on Jenkins

SSH into your Jenkins server and run:

java -version

mvn -version

git --version



2. Configure Maven in Jenkins

Go to:

Jenkins Dashboard

↓

Manage Jenkins

↓

Tools

Find:

Maven installations

Click:

Add Maven

Configure:

Name: Maven-3.8.4

You can either allow Jenkins to install Maven automatically or point it to an existing Maven installation.

For example:

Maven installations

Name:

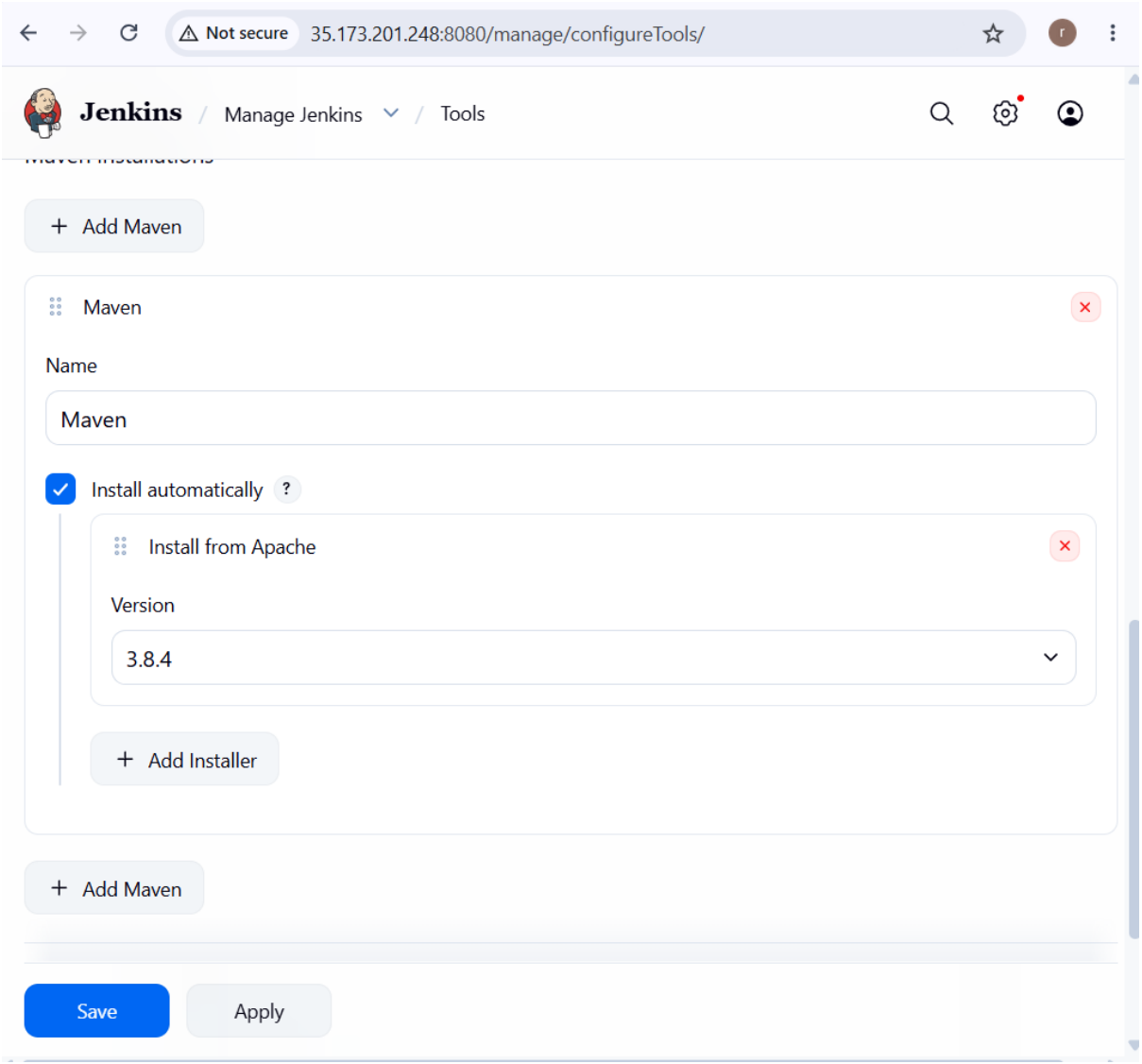
Maven-3.8.4

Version:

3.8.4

Click:

Save



3. Configure Git

Go to:

Manage Jenkins

↓

Tools

↓

Git installations

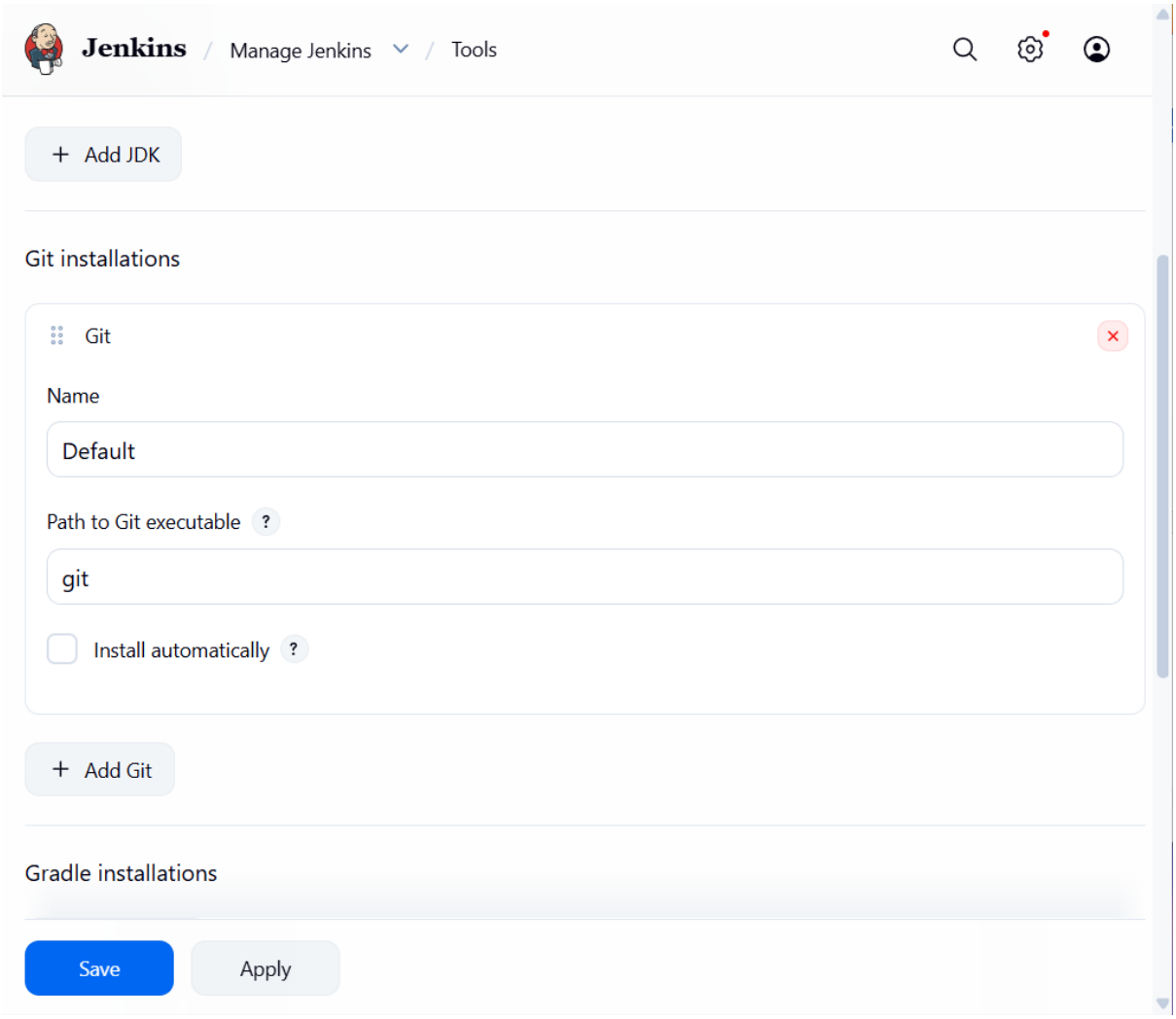
Make sure Git is configured.

For example:

Name: Default

Path to Git executable: git

If Jenkins can find Git automatically, you can leave the default configuration.



4. Create the Maven Project

Go to:

Jenkins Dashboard

↓

New Item

Enter:

Name:

java-working-app

Select:

Maven project

Then click:

OK

You now have a **Freestyle Maven Project**.

Go to:

Manage Jenkins → Plugins → Available plugins

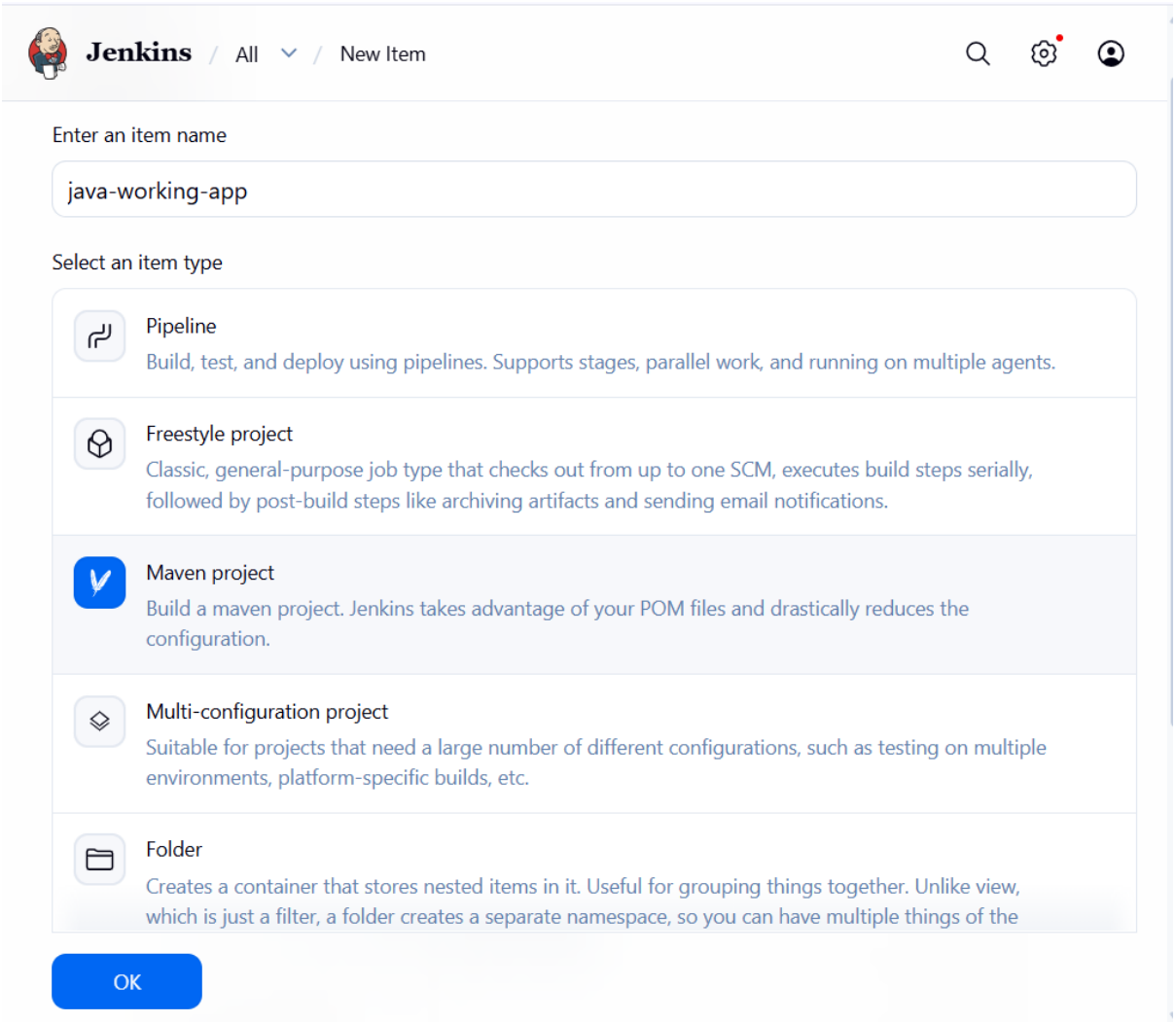
Search for:

Maven Integration

Install:

Maven Integration plugin

Then **restart Jenkins**.



5. Stage 1 — Git Clone

In the job configuration, find:

Source Code Management

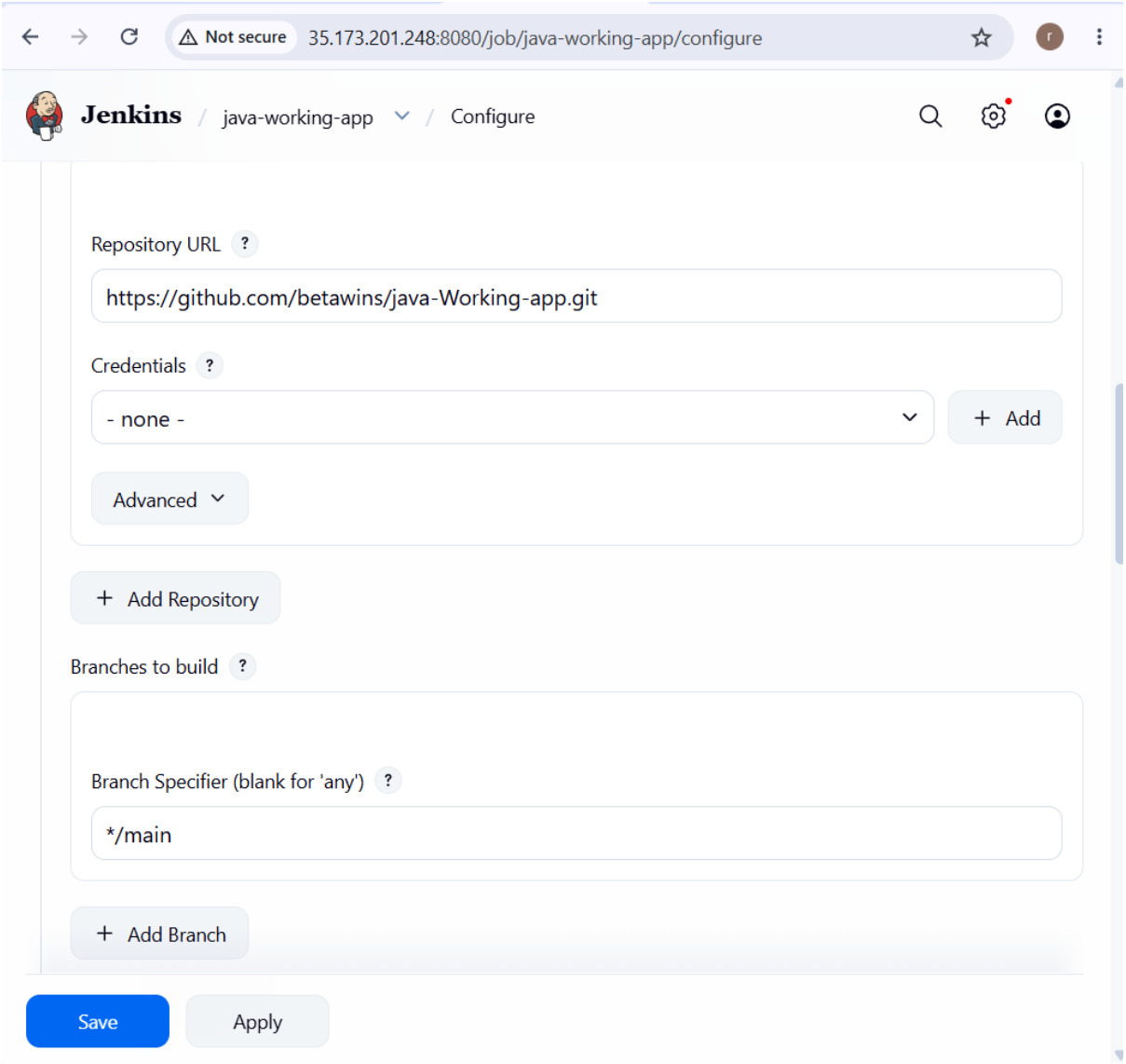
Select:

Git

Enter:

Repository URL:

https://github.com/betawins/java-Working-app.git



6. Stage 2 — Maven Compilation

Now scroll down to:

Build

Because you selected **Maven project**, Jenkins provides Maven build configuration.

Configure:

Root POM:

pom.xml

Then:

Goals and options:

clean compile

So your configuration should look like:

Root POM:

pom.xml

Goals and options:

clean compile

What does this do?

mvn clean compile

clean:

Deletes the previous target directory

compile:

Compiles Java source code

So:

pom.xml

↓

Maven

↓

Download dependencies

↓

Compile Java source code

↓

target/classes

7. Save the Job

Click:

Save

Then click:

Build Now

8. Expected Jenkins execution

You should see something similar to:

Started by user Rajkumari

Building in workspace:

/var/lib/jenkins/workspace/java-working-app

[Git] Cloning repository

https://github.com/betawins/java-Working-app.git

Checking out Revision xxxxx

[INFO] Scanning for projects...

[INFO] --- maven-clean-plugin ---

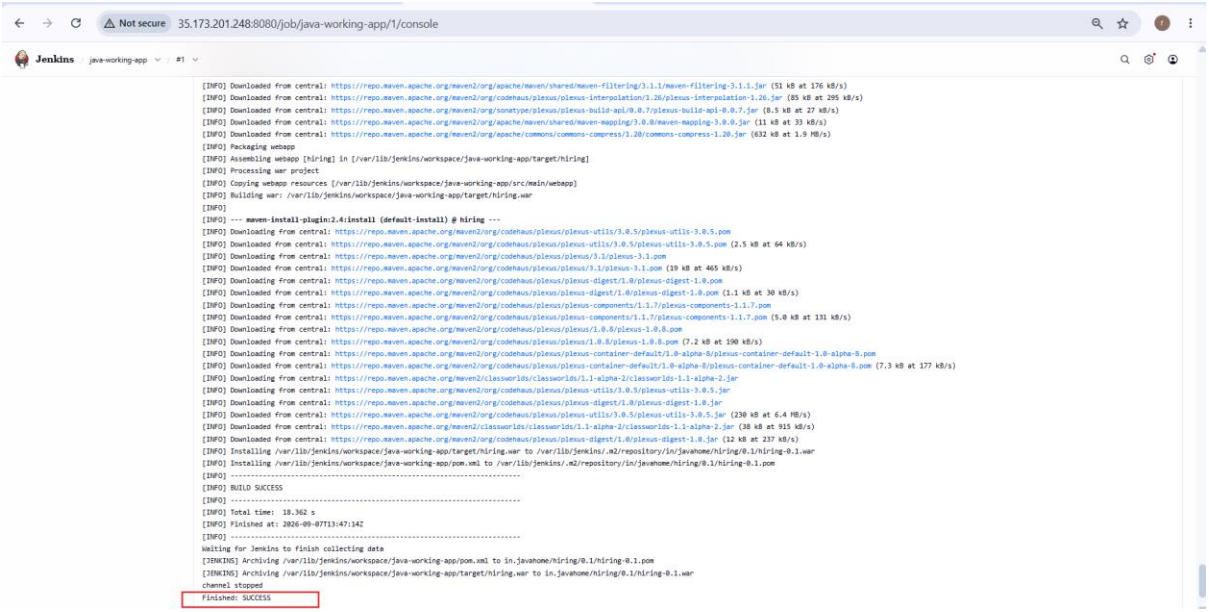
[INFO] Deleting target

[INFO] --- maven-compiler-plugin ---

[INFO] Compiling Java sources

[INFO] BUILD SUCCESS

Finished: SUCCESS



4.Use the same code and create a parameterized job in Jenkins with:

- **Stage 1: Git clone**
- **Stage 2: Maven CompilationCode: https://github.com/betawins/java-Working-app.git**

1. Create the job

Go to:

Jenkins Dashboard

↓

New Item

Enter:

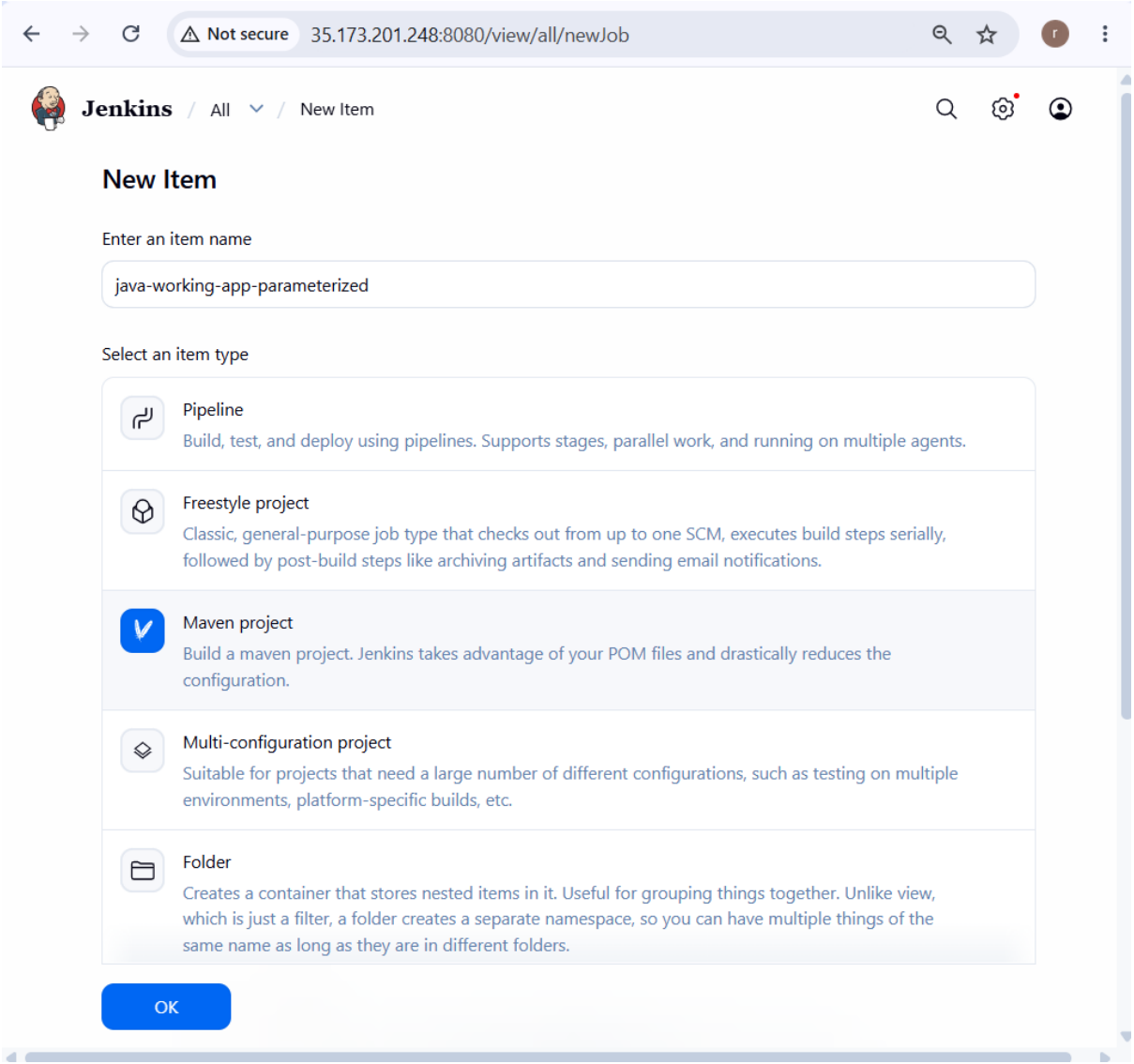
Name:

java-working-app-parameterized

Select:

Maven project

Click **OK**.



2. Enable parameterization

In the job configuration, find:

General

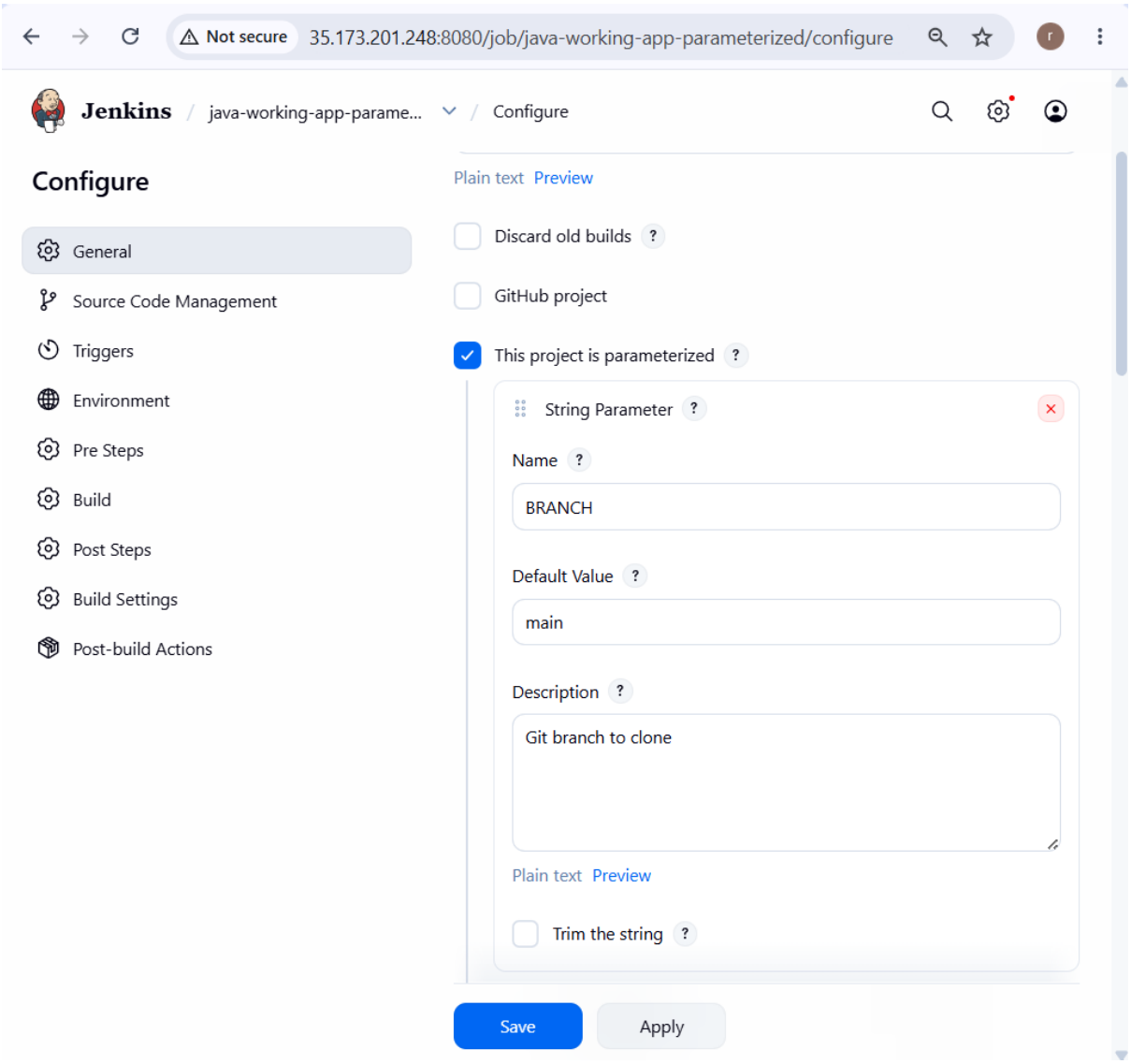
Select:

☑ This project is parameterized

Now click:

Add Parameter

We will create a **String Parameter** for the Git branch.



4. Create MAVEN_GOAL parameter

Again:

Add Parameter

↓

String Parameter

Configure:

Name:

MAVEN_GOAL

Default Value:

clean compile

Description:

Maven goals to execute

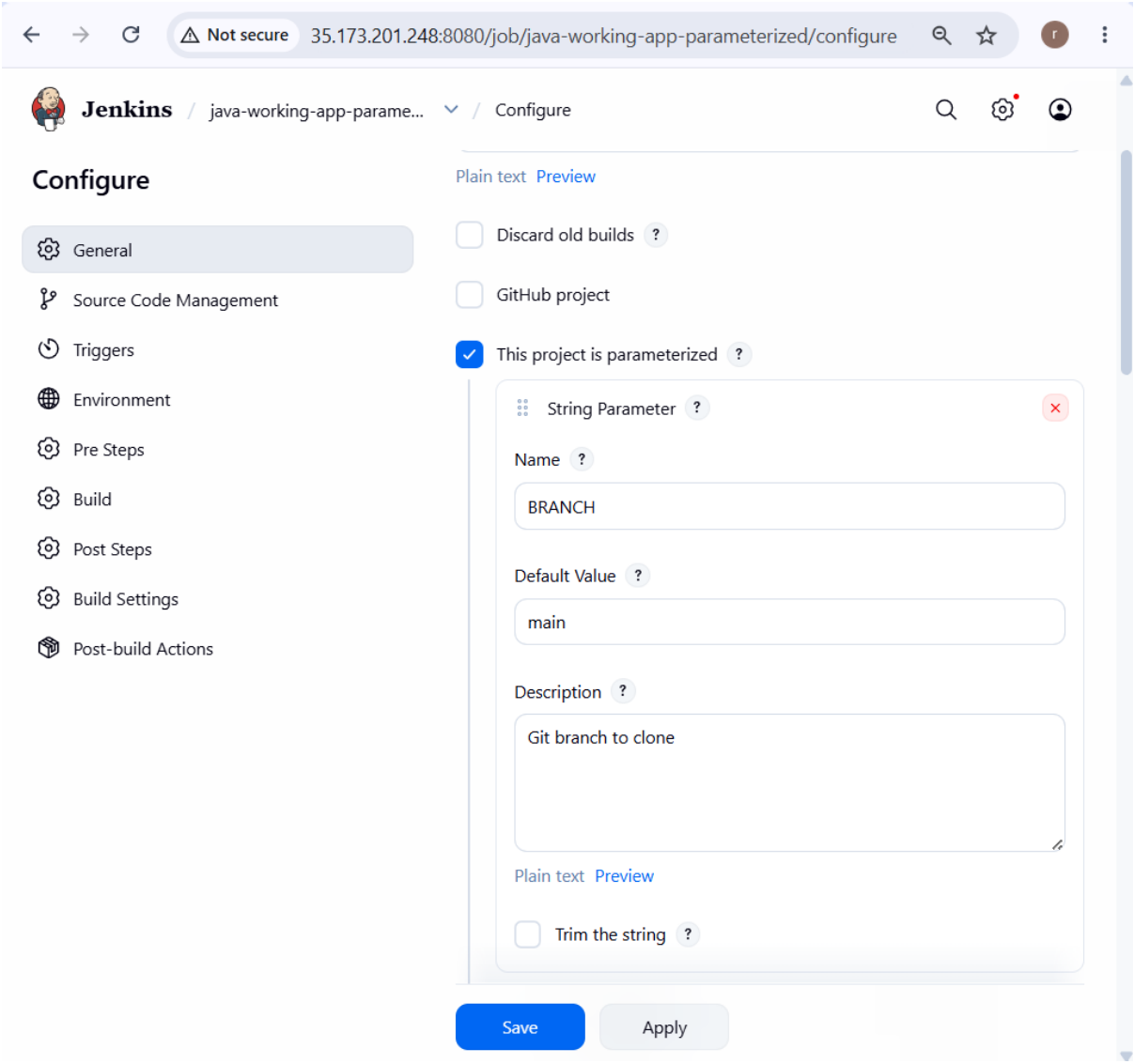
Now you have:

BRANCH

main

MAVEN_GOAL

clean compile



5. Configure Stage 1 — Git Clone

Go to:

Source Code Management

Select:

Git

Repository URL:

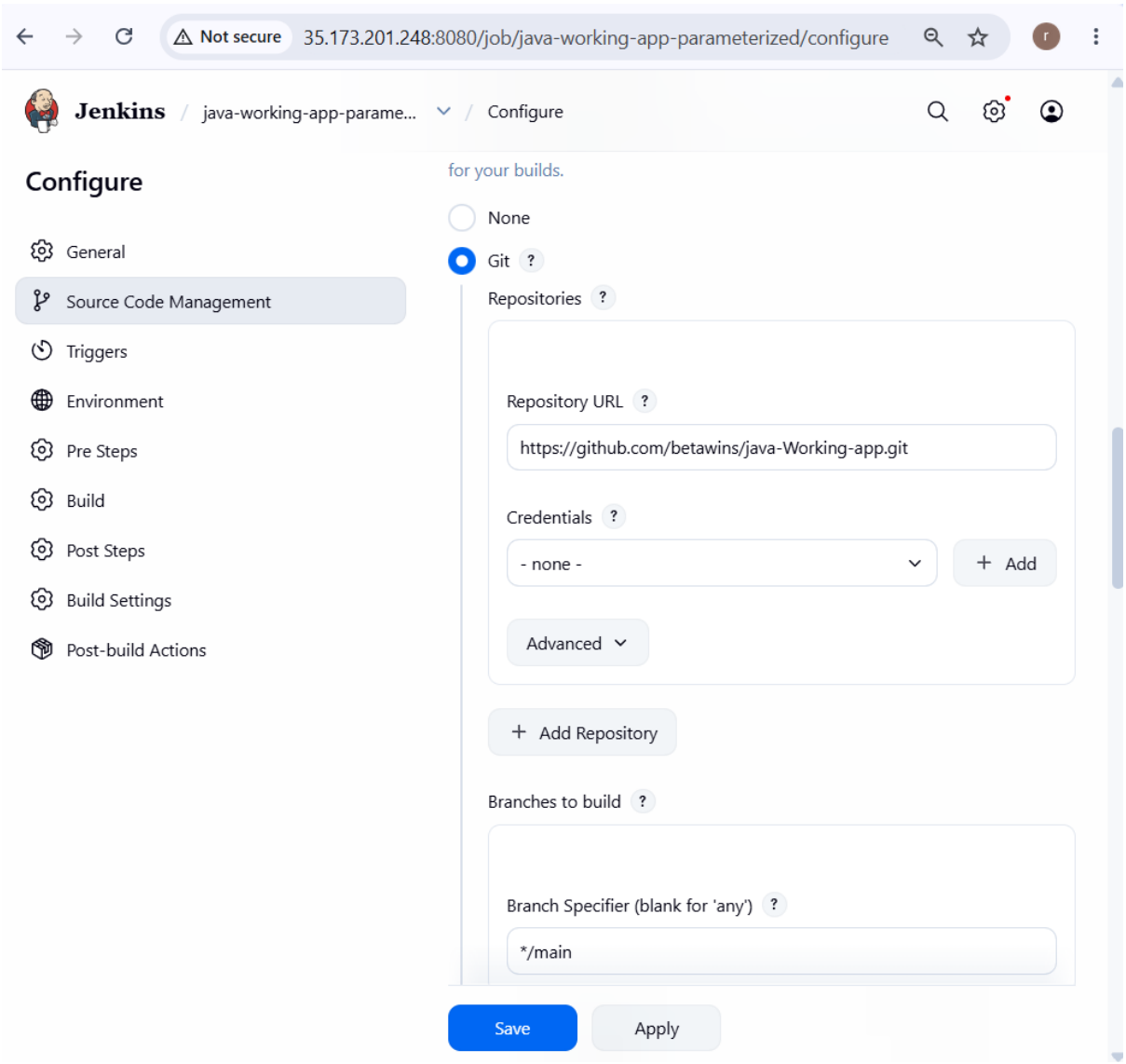
https://github.com/betawins/java-Working-app.git

For **Branch Specifier**, instead of hardcoding:

*/main

use:

*/${BRANCH}



6. Configure Stage 2 — Maven Compilation

Go to:

Build

Because this is a **Maven project**, configure:

Root POM

pom.xml

Goals and options

Use:

${MAVEN_GOAL}

So:

Root POM:

pom.xml

Goals and options:

${MAVEN_GOAL}

When you select the default:

MAVEN_GOAL = clean compile

Jenkins executes:

mvn clean compile

6. Configure Stage 2 — Maven Compilation

Go to:

Build

Because this is a **Maven project**, configure:

Root POM

pom.xml

Goals and options

Use:

${MAVEN_GOAL}

So:

Root POM:

pom.xml

Goals and options:

${MAVEN_GOAL}

When you select the default:

MAVEN_GOAL = clean compile

Jenkins executes:

mvn clean compile

7. Configure Maven installation

Make sure Jenkins has Maven configured.

Go to:

Manage Jenkins

↓

Tools

↓

Maven installations

For example:

Name:

Maven-3.8.4

Then in your Maven project configuration select:

Maven Version:

Maven-3.8.4

8. Save

Click:

Save

Now instead of **Build Now**, you should see:

Build with Parameters

Click it.

9. Run with parameters

You should see:

Build with Parameters

BRANCH

```
┌──────────────────────────────────────────────────────┐
│ main                                                 │
└──────────────────────────────────────────────────────┘
```

MAVEN_GOAL

```
┌──────────────────────────────────────────────────────┐
│ clean compile                                        │
└──────────────────────────────────────────────────────┘
```

[Build]

Leave the defaults:

BRANCH = main

MAVEN_GOAL = clean compile

Click:

Build

10. Jenkins execution

Stage 1 — Git Clone

Jenkins effectively performs:

Repository:

https://github.com/betawins/java-Working-app.git

Branch:

main

and checks the code out into:

/var/lib/jenkins/workspace/java-working-app-parameterized

Stage 2 — Maven Compilation

Jenkins reads:

pom.xml

and executes:

mvn clean compile

The flow is:

pom.xml

↓

Maven

↓

clean

↓

compile

↓

Java source files

↓

target/classes

If everything is successful:

BUILD SUCCESS

11. Try changing the parameter

Now click:

Build with Parameters

You can change:

BRANCH = development

and:

MAVEN_GOAL = clean package

Then Jenkins will execute approximately:

Git Clone

↓

development branch

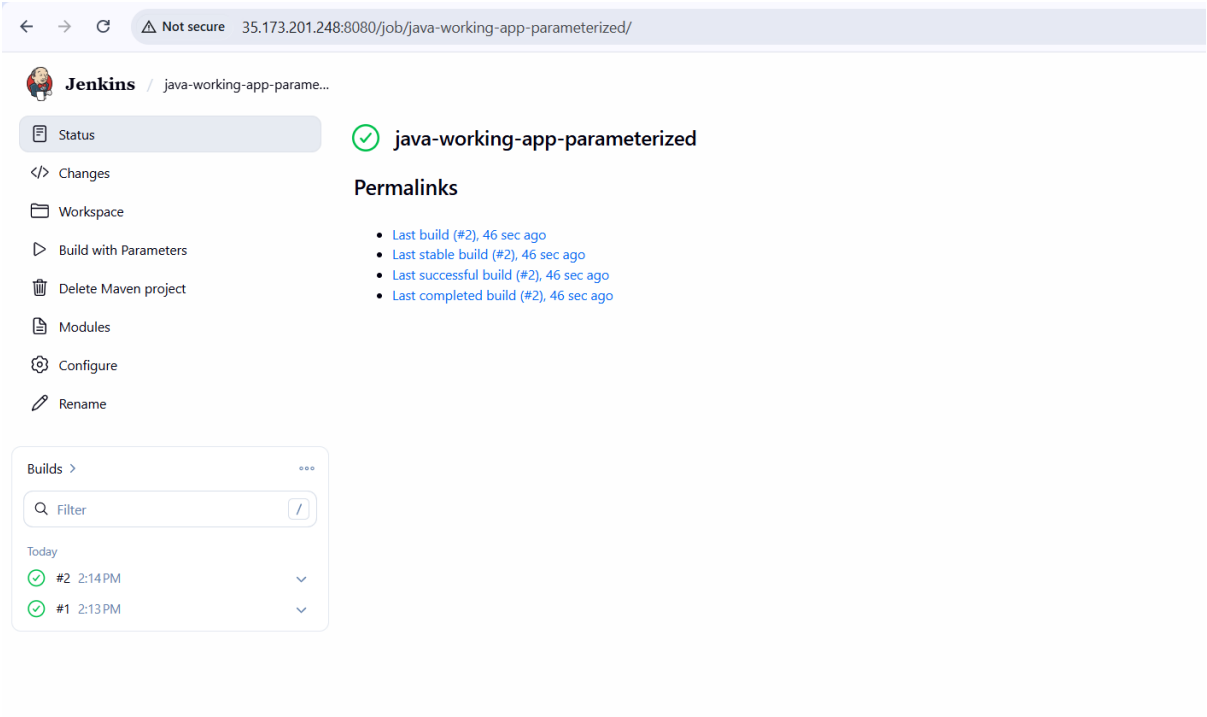
↓

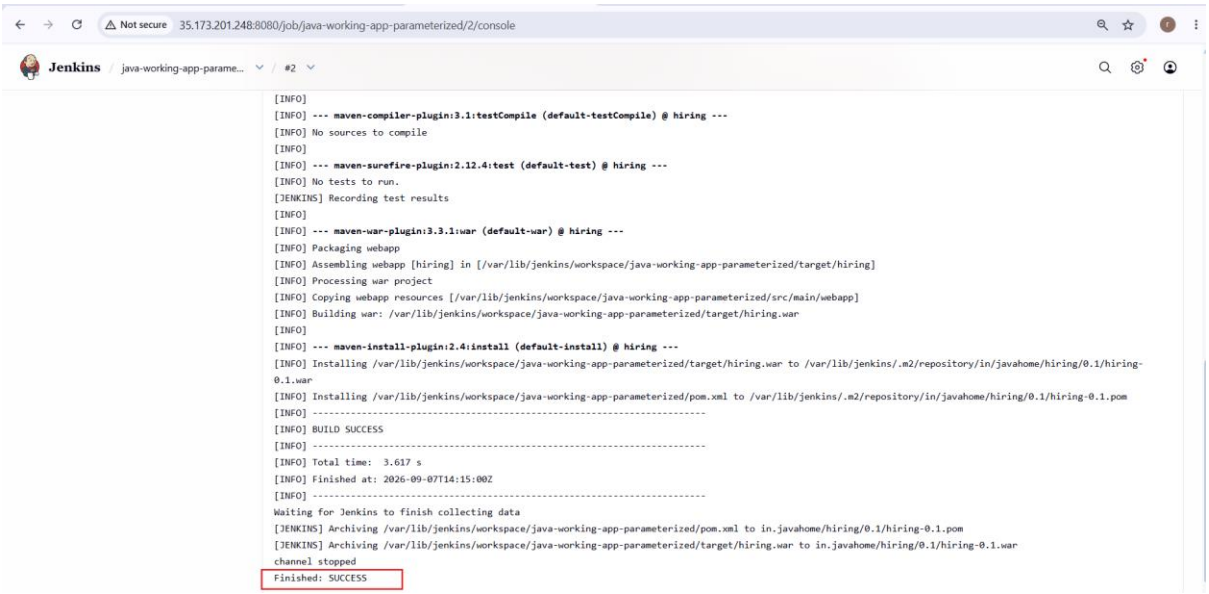
Maven

↓

mvn clean package

So the same Jenkins job can be reused for different branches and Maven operations.





5.What are the global variables in Jenkins?

In Jenkins, **global variables** are variables that are available across Jenkins jobs or throughout a Pipeline, depending on how they are defined.

There are two common meanings of "global variables" in Jenkins:

1. Built-in Jenkins environment variables

Jenkins automatically provides many environment variables.

Examples:

Variable	Meaning	Example
BUILD_NUMBER	Current build number	25
BUILD_ID	ID of the current build	25
BUILD_URL	URL of the build	http://jenkins:8080/job/demo/25/

Variable	Meaning	Example
JOB_NAME	Name of the Jenkins job	java-working-app
WORKSPACE	Workspace directory	/var/lib/jenkins/workspace/java-working-app
NODE_NAME	Jenkins node/agent name	built-in
JENKINS_HOME	Jenkins home directory	/var/lib/jenkins
GIT_COMMIT	Git commit checked out	a82f91...
GIT_BRANCH	Git branch	origin/main

You can see them in a Pipeline using:

```
pipeline {
   agent any

   stages {
      stage('Environment') {
         steps {
            sh 'echo "Job: $JOB_NAME"'
            sh 'echo "Build: $BUILD_NUMBER"'
            sh 'echo "Workspace: $WORKSPACE"'
         }
      }
```

```
    }
}
```

---

2. User-defined global variables

You can also define variables that you want Jenkins to make available broadly.

For example:

JAVA_HOME

MAVEN_HOME

TOMCAT_HOME

These can be configured through Jenkins or the operating-system environment.

For example:

MAVEN_HOME=/opt/maven

JAVA_HOME=/usr/lib/jvm/java-17

Then a Jenkins job can use them.

Environment variables vs global variables

A useful way to remember this:

```
Jenkins
  |
  +-- Built-in environment variables
  |      |
  |      +-- BUILD_NUMBER
  |      +-- JOB_NAME
  |      +-- WORKSPACE
  |      +-- BUILD_URL
```

|

+-- User-defined variables

|

+-- JAVA_HOME

+-- MAVEN_HOME

+-- TOMCAT_HOME

In your Maven job

For your java-Working-app project, you could use:

JOB_NAME

BUILD_NUMBER

WORKSPACE

GIT_BRANCH

GIT_COMMIT

For example:

echo "Job Name: $JOB_NAME"

echo "Build Number: $BUILD_NUMBER"

echo "Workspace: $WORKSPACE"

echo "Git Branch: $GIT_BRANCH"

Global variables in Jenkins are variables that provide information or configuration values that can be accessed by Jenkins jobs or Pipeline executions. Jenkins provides built-in environment variables such as BUILD_NUMBER, JOB_NAME, and WORKSPACE, and administrators can also define custom environment variables for use across jobs.